# Ilie Sarpe

*Curriculum Vitae*

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## About me

I am a researcher in algorithmic data analysis. I recently completed a postdoctoral fellowship mentored by Aristides Gionis. I got my PhD in Computer Engineering under the supervision of Fabio Vandin. My research interests focus the design of efficient, rigorous and scalable algorithms for knowledge extraction from data. I am focused on designing algorithms with *provable* theoretical guarantees, often using techniques from probability, and concentration theory.

## Education

**PhD**
*Oct 2019 - Mar 2023*
University of Padova, Italy
Department of Information Engineering

Thesis Title: Efficient and Rigorous Algorithms for the Analysis of Large Temporal Networks
Advisor: Fabio Vandin

**M.S. in Computer Engineering**
*Oct 2017 - Sep 2019*
University of Padova, Italy
Department of Information Engineering

Thesis title: Mining Motifs in Temporal Networks
Grade: 110/110 e lode (*summa cum laude*)
Advisor: Fabio Vandin

**B.S. in Computer Engineering**
*Oct 2014 - Sep 2017*
University of Padova, Italy
Department of Information Engineering

Thesis title: Statistical Correlation between Alignment-free and Edit Distance Measures
Grade: 104/110
Advisor: Matteo Comin

## Academic appointments

Postdoctoral researcher
Aug 2023 - Mar 2026 — Department of Computer Science, KTH, Stockholm (Sweden), mentored by Aristides Gionis

Visiting researcher
Oct 2024, Apr 2025 — Department of Computer Science, Brown University, Providence (RI, US), hosted by Eliezer Upfal

Research fellow
Mar 2023 - Jun 2023 — Department of Information Engineering, Unipd, Padova (Italy), mentored by Fabio Vandin

Visiting PhD student
Oct 2022 - Jan 2023 — Department of Computer Science, KTH, Stockholm (Sweden), hosted by Aristides Gionis

## Teaching

Teaching assistant
2018 - 2021 — *Big Data Computing*, Department of Information Engineering, University of Padova
2017 - 2019 — *Tutorato formativo*, Department of Information Engineering, University of Padova

## Publications

- *Ilie Sarpe*, Federico Altieri, Andrea Pietracaprina, Geppino Pucci, and Fabio Vandin, *Scalable and Distributed Silhouette Approximation*, arXiv.
- Giorgio Venturin*, *Ilie Sarpe** and Fabio Vandin, *Efficient Approximate Temporal Triangle Counting in Streaming with Predictions*, Accepted at the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD 2025). (Acceptance rate $\approx$ 24%)

- *Ilie Sarpe*† and Aristides Gionis, *Efficient and Adaptive Estimation of Local Triadic Coefficients*, Accepted at the 51st International Conference on Very Large Data Bases (VLDB 2025)
- Guangyi Zhang, *Ilie Sarpe* and Aristides Gionis, *Efficient and Practical Approximation Algorithms for Advertising in Content Feeds*, Accepted at the ACM The Web Conference 2025 (WWW 2025). (Acceptance rate 19.8%)
- *Ilie Sarpe*, Fabio Vandin and Aristides Gionis, *Scalable Temporal Motif Densest Subnetwork Discovery*, Accepted at the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2024). (Acceptance rate ≈ 20%)
- Diego Santoro* and *Ilie Sarpe*∗†, ONBRA*: Rigorous Estimation of the Temporal Betweenness Centrality in Temporal Networks*, Accepted at the ACM The Web Conference 2022 (WWW 2022). (Acceptance rate 17.7%)
- *Ilie Sarpe* and Fabio Vandin, ODEN*: Simultaneous Approximation of Multiple Motif Counts in Large Temporal Networks*, Accepted at the 30th ACM International Conference on Information and Knowledge Management (CIKM 2021). ***Selected oral presentation*** (only a small number of works were selected for presentation). (Acceptance rate 21.7%)
- *Ilie Sarpe* and Fabio Vandin, *PRESTO: Simple and Scalable Sampling Techniques for the Rigorous Approximation of Temporal Motif Counts*, Accepted at the 2021 SIAM International Conference on Data Mining (SDM21). (Acceptance rate 21.15%)

∗ denotes equal contribution, † denotes contact author, ‡ denotes alphabetical order

## Doctoral thesis

- *Efficient and Rigorous Algorithms for the Analysis of Large Temporal Networks*, 2023.

## Talks at international conferences

15th September 2025 — *Efficient Approximate Temporal Triangle Counting in Streaming with Predictions*. 22nd International Workshop on Mining and Learning with Graphs at ECMLPKDD 2025

4th September 2025 — *Efficient and Adaptive Estimation of Local Triadic Coefficients*. 51st International Conference on Very Large Data Bases London, United Kingdom - 1-5 September 2025, London, United Kingdom

27th August 2024 — *Scalable Temporal Motif Densest Subnetwork Discovery*. 2024 ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 24), 25 - 29 August 2024, Barcelona (Spain)

28th April 2022 — ONBRA*: Rigorous Estimation of the Temporal Betweenness Centrality in Temporal Networks*. ACM The Web Conference 2022 (WWW 2022), 25 - 29 April 2022, Lyon (France), Virtual Event

4th November 2021 **Selected talk** — ODEN*: Simultaneous Approximation of Multiple Motif Counts in Large Temporal Networks*. 30th ACM International Conference on Information and Knowledge Management (CIKM 2021), 1 - 5 November 2021, Queensland (Australia), Virtual Event

29th April 2021 — *PRESTO: Simple and Scalable Sampling Techniques for the Rigorous Approximation of Temporal Motif Counts*. 2021 SIAM International Conference on Data Mining (SDM 21), April 29 - May 1, 2021, Virtual Event

## Tutorials

3rd May 2025 — Aristides Gionis, Lutz Oettershagen and *Ilie Sarpe*. *Mining Temporal Graphs: Algorithms and Applications*. 2025 SIAM International Conference on Data Mining (SDM25), 1 - 3 May, 2025, Alexandria, VI, US. Lecture-style

13th May 2024 — Aristides Gionis, Lutz Oettershagen and *Ilie Sarpe*. *Mining Temporal Networks*. 2024 ACM on Web Conference 2024 (The Web Conference 2024), 13 - 17 May, 2024, Singapore. Lecture-style

## Invited talks

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| 10th March 2026 | *Efficient and Adaptive Estimation of Local Triadic Coefficients*. National Institute of Informatics (NII), seminar |
| 23rd December 2025 | *Optimal and Efficient Link Insertion for Hitting-Time Minimization*. Department of Information Engineering, University of Padova, seminar |
| 8th September 2025 | *Efficient and Adaptive Estimation of Local Triadic Coefficients*. University of Liverpool. Algorithms, Complexity Theory and Optimisation Series, seminar |
| 7th July 2025 | *Scalable Temporal Motif Densest Subnetwork Discovery*. Algorithmic Aspects of Temporal Graphs VIII. Workshop at ICALP 2025 |
| 3rd September 2024 | *Adaptive Estimation of Triadic Coefficients*. Statistical and Probabilistic Methods in Algorithmic Data Analysis, Dagsthul seminar |
| 13th September 2023 | *Discovering Temporal Motif Densest Subnetworks*. Workshop on Algorithmic Aspects of Clustering and Related Problems (ALACARTE 2023), Bertinoro BICI |
| 22nd December 2021 | *Motifs in Temporal Networks Definitions, Algorithms and Applications*. Invited lecture for the *Learning from Networks* M.Sc. course, Department of Information Engineering, University of Padova, Italy |

## Fellowships and awards

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| Jun 2025 | Outstanding reviewer for KDD2025 (second cycle). |
| Sep 2023 | Bertinoro BICI travel fellowship. |
| Oct 2022 | SoBigData++ Transnational Access (known as TNA) support for short term visits. |
| Oct 2019 - Dec 2022 | PhD Fellowship from "Department of Information Engineering (DEI)", University of Padova, Italy |
| 2017 | Award for scientific degrees, award given to the best 500 students of scientific degrees, University of Padova, Italy |
| 2017 | "Mille e una lode", award for the top 5% students of the academic year 2016, University of Padova, Italy |
| 2016 | "Mille e una lode", award for the top 5% students of the academic year 2015, University of Padova, Italy |

## Student supervision

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| PhD students | Giorgio Venturin (University of Padova, 2023 - current, co-supervised) |
| Master students | Davide Peressoni (University of Padova, 2022, co-supervised)<br>Giorgio Venturin (University of Padova, 2023, co-supervised)<br>Tommaso Calvani (Sapienza university, 2025, co-supervised)<br>Theofanis Georgakopoulos (KTH university, 2025, co-supervised) |
| Bachelor students | Filippo Ronco (University of Padova, 2020, co-supervised)<br>Vidyaratnam Ganapathy (KTH university and Purdue, 2025, co-supervised) |

## Service

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| Area chair | KDD 2026 (second cycle) |
| Program committee member | KDD 2023, ECCB 2023, WSDM 2024, SDM 2024, WWW 2024, KDD 2024, ACML 2024, WSDM 2025, KDD 2025 (first cycle), KDD 2025 (second cycle, **outstanding reviewer**), WWW 2025, GrAPL 2025 (IPDPS2025 workshop), CIKM 2025, TGL 2025 (KDD 2025 workshop), MLG 2025 (ECML-PKDD2025 workshop), KDD 2026 (first cycle), WWW 2026, WSDM 2026 (demo track), VLDB 2026, ICML 2026, Euro-Par 2026 (Theory and Algorithms track), NeurIPS 2026, ACM AI Summit 2026 |
| Conference reviewer | RECOMB 2020, KDD 2020, ICDM 2020, WWW 2021, ECML-PKDD 2021, ICDM 2021, WSDM 2022, WWW 2022, KDD 2022, ICDM 2022, ACDA23, WWW 2023, ECML-PKDD 2023, ESA 2023, ISAAC 2024, SDM 2025, SWAT 2026, ESA 2026 (S-track) |
| Journal reviewer | Journal of Graph Algorithms and Applications, Data Mining and Knowledge Discovery, Knowledge and Information Systems, Information Processing and Management |
| Session chair | TheWebConference 2024, Session: Algorithms for social networks |
| PhD admission | Committee member at KTH in the group of Prof. Aristides Gionis |

## Funding

Researcher

PRIN Project n. 20174LF3T8 AHeAD (Efficient Algorithms for Harnessing Networked Data), MIUR Italy

"SID 2020: RATED-X", University of Padova, Italy

ERC Advanced Grant REBOUND n. 834862

EC H2020 RIA project SoBigData++ n. 871042

## Software packages

- PRESTO: an efficient sampling algorithm for estimating the count of a temporal motif in a temporal network. `https://github.com/VandinLab/PRESTO`
- ODEN: and efficient sampling algorithm for estimating the counts of multiple temporal motifs sharing a common topological structure. `https://github.com/VandinLab/odeN`
- ONBRA: an efficient algorithm for estimating the temporal betweenness centrality of the various nodes in a temporal network under two criteria for the paths considered. `https://github.com/iliesarpe/onbra`
- ALDENTE: a suite of efficient and scalable algorithms to extract dense subnetworks, where density accounts for the number of the temporal motif occurrences. `https://github.com/iliesarpe/ALDENTE`
- Triad: an efficient and adaptive algorithm to compute the average of local triadic coefficients on graph data. `https://github.com/iliesarpe/triad`
- Scalable Silhouette: a suite of scalable algorithms for computing an estimate of the silhouette of a $k$-clustering of $n$ points in subquadratic time. The suite can estimate the local silhouette of all points and the global silhouette, both single threaded and in parallel. `https://github.com/iliesarpe/ScalableSilhouetteComputation`

## Other information

Programming experience: C++, C, Java, Python, MATLAB, SQL, NoSQL, Bash, LaTeX

Programming frameworks: Apache Spark, IBM ILOG CPLEX, Gurobi

Languages: Native Italian and Romanian Speaker, B2 English